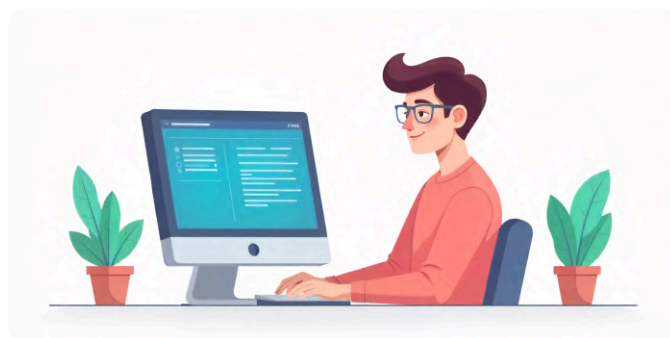


Introduction to DevOps Practices



A. WARM-UP QUESTIONS

- What is the difference between software development and operations?
- How do companies deploy software applications to users?
- What happens when a website or app goes down?
- Why do some software updates cause problems?
- How can teams work together more effectively on technology projects?

B. VOCABULARY PREVIEW

Match the words with their meanings:

- | | |
|-------------------------|--|
| 1. deployment ____ | a. system for tracking changes to code over time |
| 2. infrastructure ____ | b. combining different software components to work together |
| 3. automation ____ | c. using computers and internet services provided by other companies |
| 4. pipeline ____ | d. lightweight package containing application and its dependencies |
| 5. monitoring ____ | e. continuously watching system performance and health |
| 6. scaling ____ | f. the basic technical systems that support an application |
| 7. container ____ | g. process of releasing software to users |
| 8. cloud computing ____ | h. percentage of time a system is working correctly |
| 9. version control ____ | i. using technology to perform tasks without human intervention |
| 10. integration ____ | j. increasing system capacity to handle more users or data |
| 11. reliability ____ | k. how dependable and consistent a system performs |
| 12. uptime ____ | l. series of automated steps for processing code changes |

Meet Alex Chen: DevOps Engineer and System Architect

Alex Chen works as a DevOps Engineer at a fast-growing e-commerce company called TechMart. Her primary responsibility is ensuring that the company's website and mobile applications run smoothly, **scale** efficiently, and deploy new features without interrupting service for millions of customers worldwide.

Before Alex joined the company, the development and operations teams worked separately, often causing conflicts and delays. Developers would write code and "throw it over the wall" to operations staff, who were responsible for running it in production. When problems occurred, both teams would blame each other, and customers suffered from frequent outages and slow performance.

DEVOPS INDUSTRY FACTS

The DevOps market is expected to reach \$12.85 billion by 2025, growing at 18.2% annually. Companies adopting DevOps practices deploy code 200 times more frequently than traditional organizations, with 2,555 times faster lead times. Organizations with strong DevOps cultures report 50% fewer security failures and 22% less time spent on unplanned work.

Alex's role bridges this gap between development and operations. She works closely with software developers to understand how applications work, while also managing the **infrastructure** that runs those applications. Her goal is to create seamless processes that allow new code to move from a developer's computer to customer-facing systems quickly and safely.

One of Alex's main tasks is building and maintaining **deployment pipelines**. These automated systems take code written by developers, test it thoroughly, and deploy it to production servers. Before implementing these pipelines, manual deployments took hours and often contained errors. Now, the automated process completes deployments in minutes with much higher **reliability**.

Container technology plays a crucial role in Alex's daily work. She packages applications into containers—lightweight, portable units that include everything needed to run the application. Containers ensure that software behaves consistently whether running on a developer's laptop, testing servers, or production systems. This consistency eliminates many problems caused by environmental differences.

Cloud computing has revolutionized how Alex manages infrastructure. Instead of purchasing and maintaining physical servers, TechMart uses cloud services from Amazon Web Services (AWS). Alex can create new servers instantly, scale

resources up or down based on demand, and only pay for what the company actually uses. This flexibility is essential during peak shopping seasons when traffic increases dramatically.

Monitoring represents another critical aspect of Alex's responsibilities. She implements comprehensive monitoring systems that track application performance, server health, and user experience metrics. When problems occur, these systems automatically alert the team and often trigger automated responses to resolve issues before customers notice them.

The most rewarding aspect of Alex's job is solving complex problems that directly impact customer experience. DevOps requires a combination of technical skills and collaborative mindset. Alex must understand multiple programming languages, infrastructure technologies, and operational procedures while also communicating effectively with diverse teams.

COMPREHENSION

- What problems existed before Alex's company adopted DevOps practices?
- How do deployment pipelines improve the software release process?
- What advantages do containers provide for application deployment?
- Why is monitoring important in DevOps, and how does it help prevent problems?
- What combination of skills does Alex need to be successful in DevOps?

VOCABULARY REVIEW

Fill in the blanks:

The development team uses _____ to track all changes made to the application code.

_____ allows the company to create new servers instantly without buying physical hardware.

The automated _____ runs tests and deploys code to production servers in minutes.

_____ systems alert the team immediately when performance problems occur.

_____ ensures that applications run consistently across different environments.

GRAMMAR REVIEW - PRESENT PERFECT & MODAL VERBS FOR POSSIBILITY

Complete the sentences:

DevOps practices _____ (improve) deployment reliability significantly over the past five years.

Cloud computing _____ (enable) companies to scale their infrastructure more flexibly recently.

Automation _____ reduce human errors in deployment processes.

The monitoring system _____ detect problems before they affect customers.

Container technology _____ (solve) many environment-related deployment issues.

ADDITIONAL EXERCISES

DevOps Tool Categories

Match each tool with its primary function in the DevOps workflow:

Tools:

Docker, Jenkins, Git, Prometheus, Kubernetes, Terraform, Slack, AWS

Categories:

Version Control: _____

Containerization: _____

Container Orchestration: _____

Continuous Integration: _____

Infrastructure as Code: _____

Monitoring: _____

Communication: _____

Cloud Platform: _____

Deployment Pipeline Design

Design a basic deployment pipeline for a web application:

Pipeline Stages:

Put these stages in the correct order (1-8):

- ___ Deploy to production servers
- ___ Run automated tests
- ___ Build application packages
- ___ Developer commits code changes
- ___ Notify team of deployment status
- ___ Deploy to staging environment for final testing
- ___ Merge approved changes into main branch
- ___ Perform security and quality checks

Quality Gates:

At which stages should the pipeline stop if problems are detected?

- Stage ___: If code doesn't compile
- Stage ___: If automated tests fail
- Stage ___: If security vulnerabilities found
- Stage ___: If staging deployment fails

Match the Picture to the Term

👉 Look at the pictures. Match each picture with the correct DevOps term. Write the number of the picture next to the word.



1



2



3



4



5



6

Match the terms:

• Cloud computing: _____

• Deployment: _____

• Monitoring: _____

• Automation: _____

• Pipeline: _____

• Version control: _____

DevOps Metrics Dashboard

Design a monitoring dashboard with key performance indicators:

APPLICATION METRICS	INFRASTRUCTURE METRICS	BUSINESS IMPACT
Response Time: Target < 200ms	Server CPU: Target < 70%	System Uptime: Target 99.9%
Current: ____ms, Trend: ↑↓→, Action: Yes/No	Web servers: __%, Database: __%	Current month: __%
Error Rate: Target < 0.1%	Action needed: Yes/No	Downtime incidents: __, Revenue impact: \$__
Current: __%, Trend: ↑↓→, Action: Yes/No		

ANSWER KEYS

Match the Picture to the Term

1. Automation, 2. Pipeline, 3. Cloud computing, 4. Monitoring, 5. Deployment, 6. Version control

Vocabulary Preview

1-g, 2-f, 3-i, 4-l, 5-e, 6-j, 7-d, 8-c, 9-a, 10-b, 11-k, 12-h

Vocabulary Review

1. version control, 2. Cloud computing, 3. pipeline, 4. Monitoring, 5. Containerization

Grammar Review

1. have improved, 2. has enabled, 3. can, 4. can, 5. has solved

DevOps Tool Categories

Version Control: Git

Containerization: Docker

Container Orchestration: Kubernetes

Continuous Integration: Jenkins

Infrastructure as Code: Terraform

Monitoring: Prometheus

Communication: Slack

Cloud Platform: AWS

Deployment Pipeline Order

4, 7, 2, 1, 6, 8, 3, 5

Quality Gates

Stage 2: If code doesn't compile

Stage 3: If automated tests fail

Stage 4: If security vulnerabilities found

Stage 6: If staging deployment fails